CNN-Based Plant Disease Classification

End-to-End Deep Learning System Using the PlantVillage Dataset

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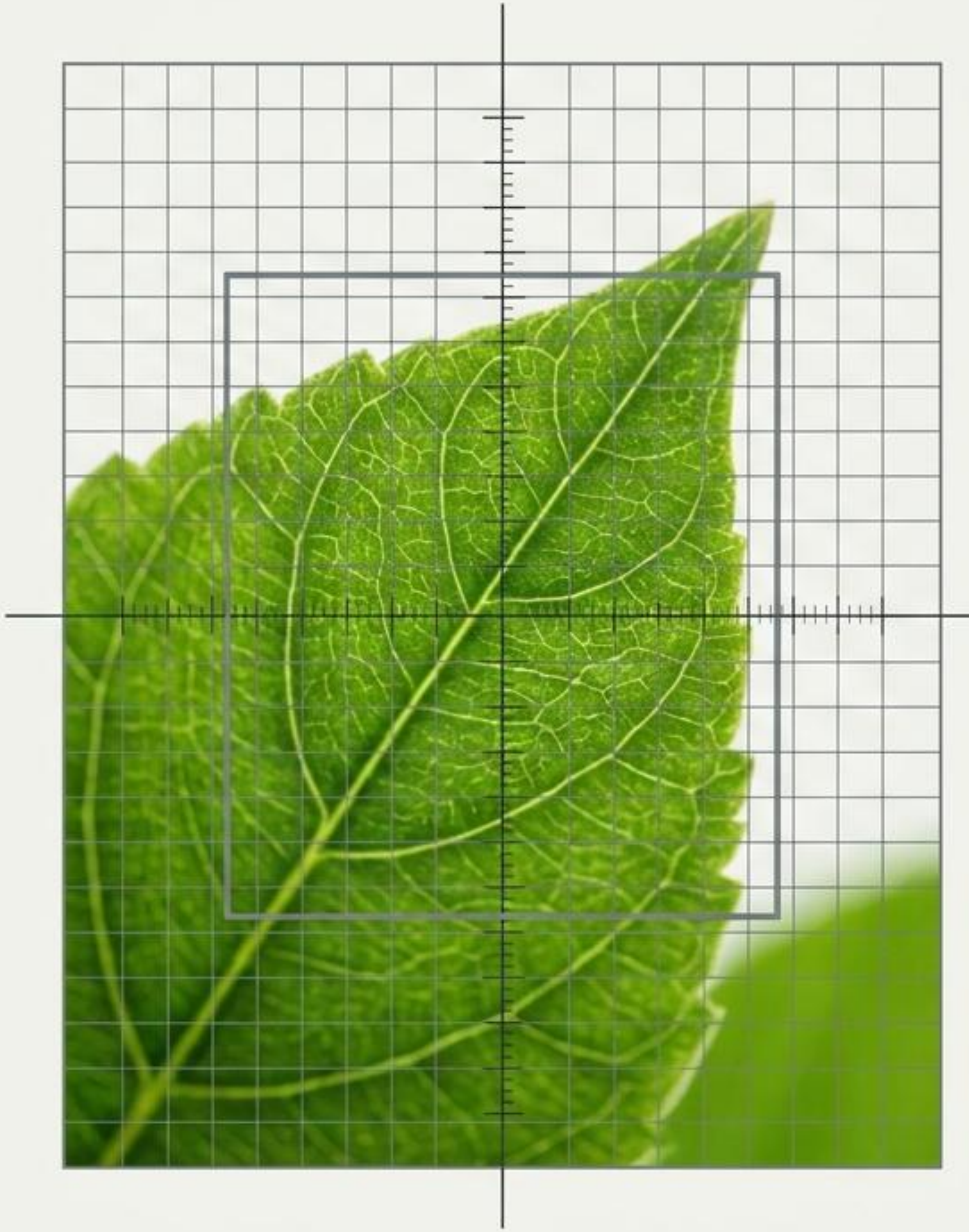


Establishing the PlantVillage Benchmark

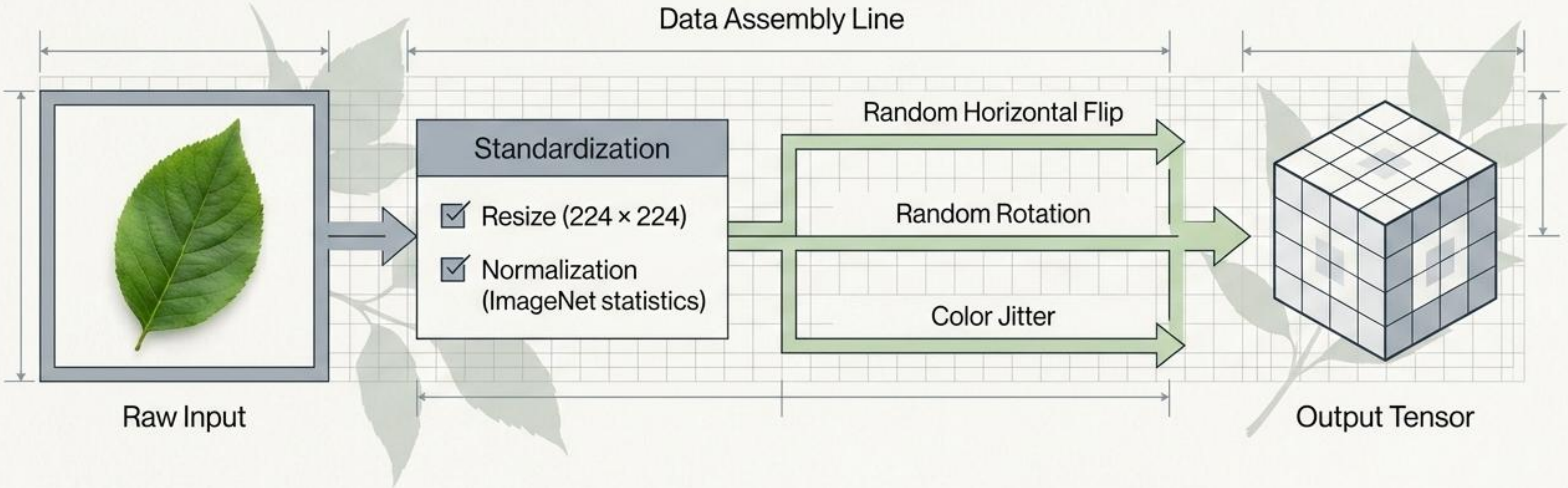
Primary Objective: Build, improve, and objectively compare multiple deep learning architectures to classify disease status from raw leaf images.

~54,000
Structured RGB leaf images

30+
Distinct plant disease and healthy classes

Directory Structure
Class-based categorization folders

The Preprocessing & Augmentation Pipeline



Split:	Optimizer:	Loss Function:	Batch Size:
Stratified Train / Validation / Test	Adam	CrossEntropyLoss	32

Week 2: Establishing a CNN Baseline

Status: Trained from Scratch

3 Convolutional Layers → Max Pooling Layers → Fully Connected Classifier

Early Successes

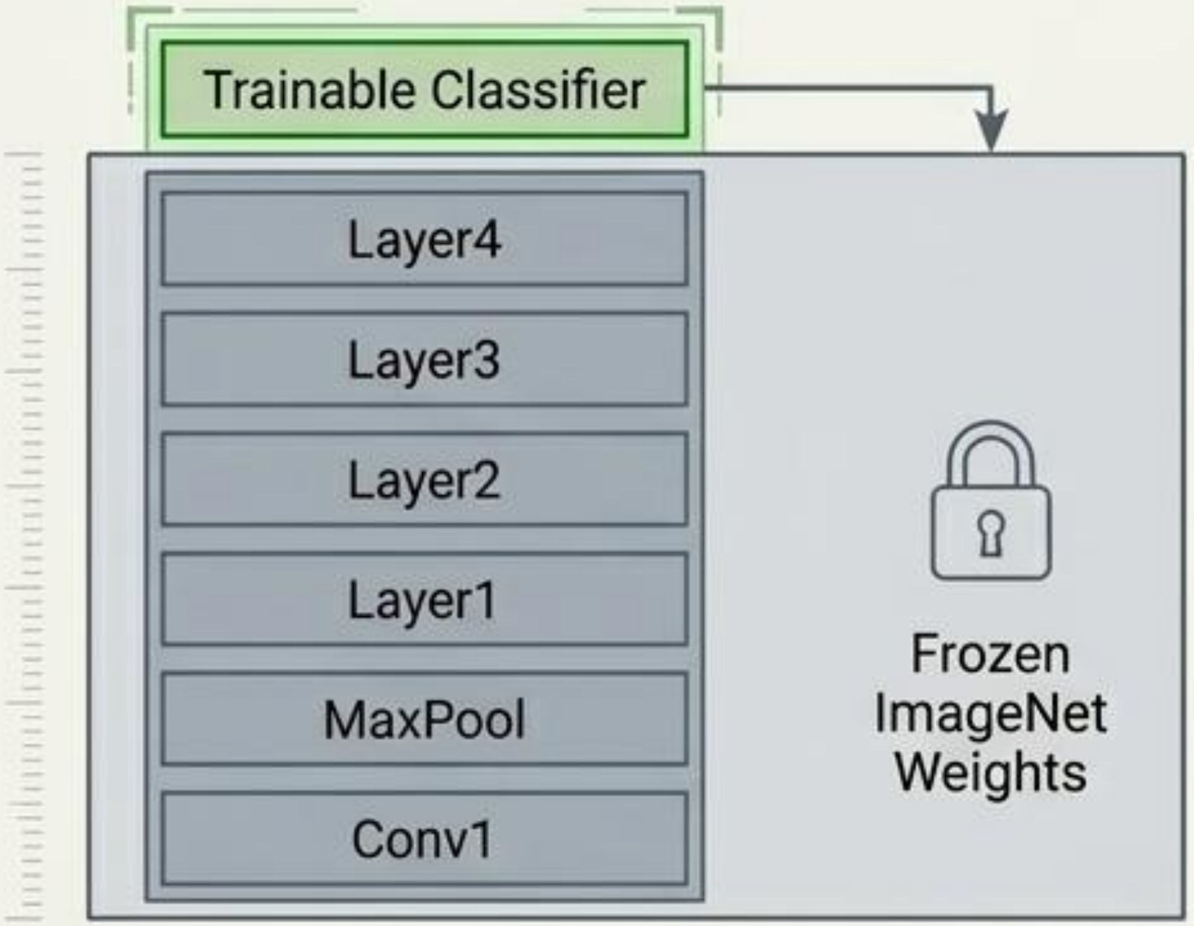
The baseline model successfully learns basic, low-level leaf features, establishing a functional proof-of-concept for the classification pipeline.

Critical Limitations

- **Limitation 1:** The model struggles significantly to differentiate between visually similar disease classes.
- **Limitation 2:** Early signs of overfitting observed due to limited architectural depth and parameter constraints.

Architectural Evolution via ResNet18

Week 3: Feature Extraction



Week 4: Final Model - Full Fine-Tuning

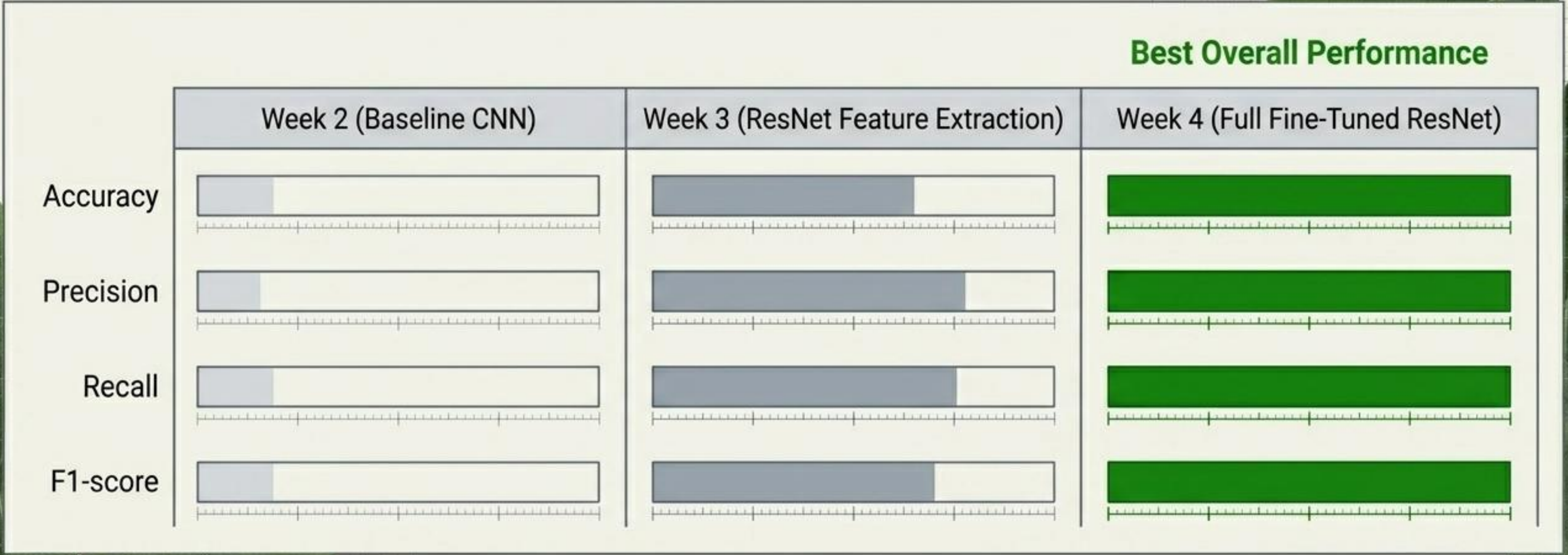
Stabilized Learning Rate: 0.0001

Learning Rate Scheduling Active

Key Engineering Insight

Shifting from basic feature extraction to full fine-tuning—combined with a strict dataset splitting strategy to prevent data leakage—resulted in massively improved training stability and the highest overall classification performance.

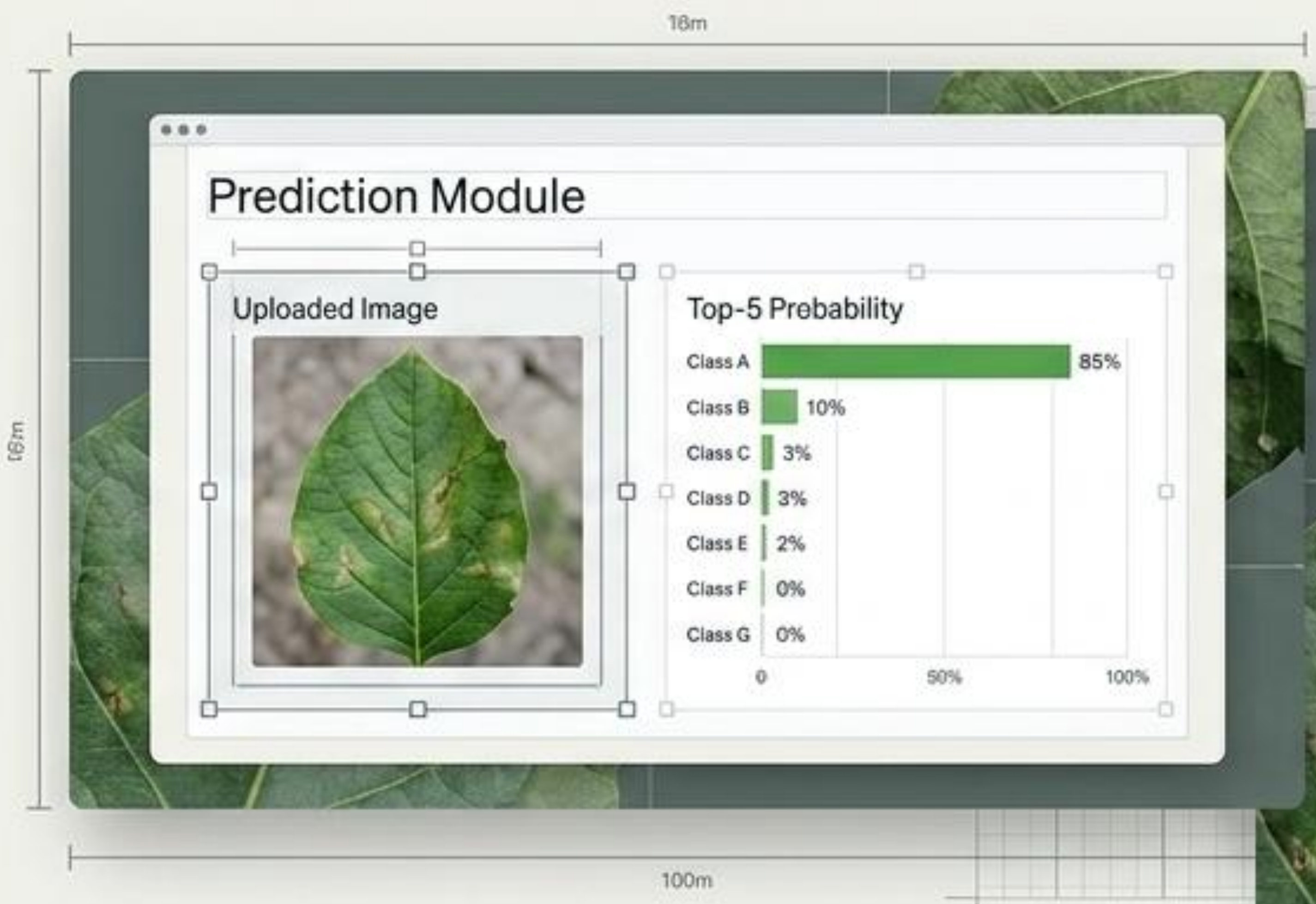
Iterative Performance Matrix

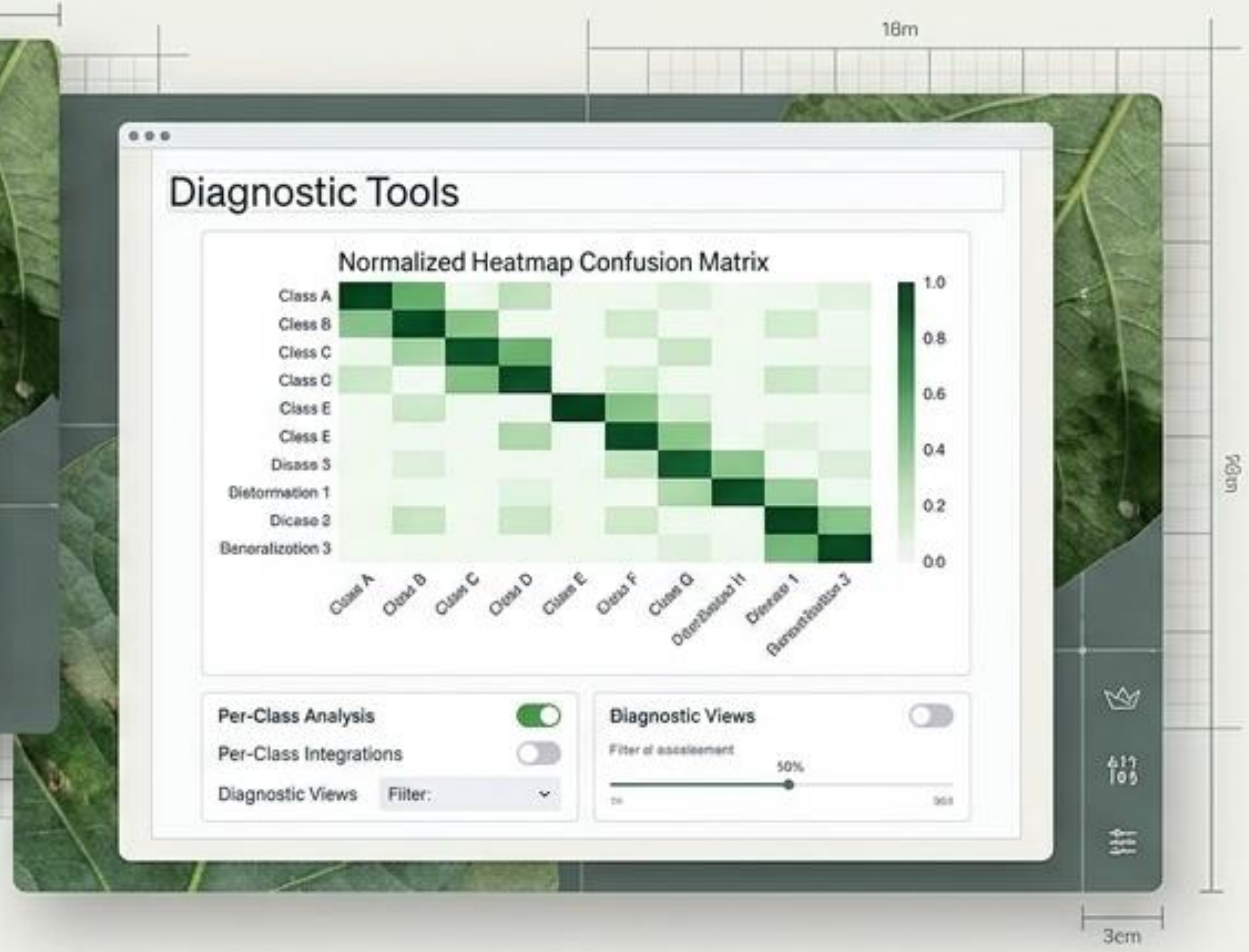


Error Analysis

Persistent Diagnostic Challenges: While the fine-tuned architecture resolved most baseline limitations, confusion remains regarding early-stage symptoms and highly visually similar leaf diseases, compounded by underlying class imbalances in the raw dataset.

Interactive Streamlit Deployment





System Workflow

1. User uploads image
2. Image passes through preprocessing pipeline
3. Input routed to CNN / ResNet models
4. Ensembled predictions displayed in dashboard

System Achievements & Future Scope

Project Achievements

- Engineered an end-to-end pipeline from raw PlantVillage data to a deployed application.
- Successfully optimized a baseline CNN into a fully fine-tuned ResNet18 architecture.
- Developed a robust comparative dashboard for real-time model inference and error analysis.

The Next Iteration

- Architecture Upgrade: Transitioning from ResNet to EfficientNet or Vision Transformers.
- Data Robustness: Moving beyond clean dataset conditions to train on real-world, noisy agricultural field datasets.
- Deployment Scaling: Migrating the backend to cloud infrastructure (HuggingFace / AWS) and developing a dedicated mobile application version for in-field use.

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